THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The energy currency of the cell is: 1

(a) ADP **(b)** glucose **(c)** starch **(d)** ATP

2. Artificial kidney works on the principle of: 1

(a) filtration only **(b)** photosynthesis **(c)** dialysis **(d)** osmosis only

3. How are fats digested in our body? The site of fat digestion is: 1

(a) small intestine **(b)** large intestine **(c)** mouth **(d)** stomach

4. Haemoglobin transports mainly: 1

(a) food **(b)** oxygen **(c)** water **(d)** nitrogen

5. Food is transported in plants through: 1

(a) stomata **(b)** phloem **(c)** xylem **(d)** cambium

6. What is the role of saliva in the digestion of food? 1

(a) it absorbs water
(b) its amylase breaks down starch into sugar
(c) it breaks down proteins
(d) it emulsifies fats

7. The gaseous exchange in humans takes place in: 1

(a) larynx **(b)** bronchi **(c)** alveoli **(d)** trachea

8. The process of loss of water in the form of vapour from aerial parts of plants is: 1

(a) transpiration **(b)** translocation **(c)** respiration **(d)** absorption

9. Which is the correct sequence of parts in the human alimentary canal? 1

(a) oesophagus -> mouth -> stomach -> large intestine -> small intestine
(b) mouth -> oesophagus -> stomach -> small intestine -> large intestine
(c) mouth -> stomach -> oesophagus -> small intestine -> large intestine
(d) mouth -> oesophagus -> stomach -> large intestine -> small intestine

10. The blood vessel that carries oxygenated blood from the lungs to the heart is: 1

(a) vena cava **(b)** aorta **(c)** pulmonary vein **(d)** pulmonary artery

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What advantage over an aquatic organism does a terrestrial organism have with regard to the availability of oxygen for respiration? 2

12. How is the amount of urine produced regulated? 2

13. What is the role of acid in our stomach? 2

14. What are the different ways in which glucose is oxidised to provide energy in various organisms? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What is the role of bile juice, pancreatic juice and intestinal juice in digestion? 3

16. Describe how water is transported in plants up to a great height. 3

17. Why is the heart divided into four chambers? 3

OR

How does food enter the stomach and move in the alimentary canal? 3

18. Explain translocation of food in plants. 3

OR

What is the importance of the nephron's tubular reabsorption? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the process of breathing in human beings and the exchange of gases at the alveoli. 5

OR

Compare aerobic respiration, anaerobic respiration in yeast, and anaerobic respiration in muscle. 5

20. Describe the structure and working of the human heart with the flow of blood. 5

OR

Explain how the structure of the lungs suits gas exchange, and how oxygen and carbon dioxide are carried in blood. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND